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4-D Geomechanics Modelling for Potential CO_2 Storage Site in Malaysia – Fault Reactivation and Maximum Pressure Injection Limit

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Abstract

This paper presents a study on the risk associated with CO_2 injection in geological storage and fault reactivation through a comprehensive workflow for determining the feasibility of CO_2 storage campaign in carbonate reservoir in Malaysia. The study includes constructing a 4-D coupled reservoir geomechanical model and developing a workflow that can be used to evaluate geomechanics risks associated with carbon capture and storage (CCS) by outlining results and findings that drive key decisions in the planning of CCS strategy.

The workflows aims to better delineate and enumerate the risks with CCS as it constructing and calibrating single well models by corroborating numerous inputs including stringent laboratory testing data and drilling analysis, and combining with structural model and reservoir model to create a field wide 4-D geomechanical model using advanced time lapsed geomechanics simulation. Coupled simulations with the dynamic reservoir model provided predictions of the fault stability by considering fault deformation. The paper further highlights the geomechanics evaluation consideration (economics and engineering trade-off) in designing maximum safe injection pressure for CO_2 sequestration program.

The results of the study show fault condition subjected to different time-steps of the coupled simulation during depletion and injection. At each time-step, the development of plastic shear strain and absolute displacement are plotted and risks associated with the change in reservoir pressure are assessed and quantified. Different injection plans are modelled to determine the impact on final storage capacity, long term fluid containment and upper safe injection limit to avoid breaching the caprock.

The study offers the utilization of the latest techniques in 4-D coupled geomechanical modelling which reduced the study time and cost significantly, making it affordable for in-time solution for decision making. The paper also aims to encourage the consideration of the applied novel workflow involved in CCS strategye valuation focusing on risk assessment which ultimately will affect reservoir maximum safe injection limit, capacity, long term storage safety, and monitoring program to mitigate potential geohazard leakage.

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Introduction

The carbonate formations in the Central Luconia province offshore Sarawak, Malaysia generally form prolific gas producing reservoirs. The field being studies is operated by PETRONAS and is located as shown in Figure 1. The field is primarily of platform type carbonate build-up with a single long gas column overlying an aquifer, and the field production started in 1995. GPS surveillance data, measured on the platform over the past 20 years, have clearly shown a number of striking events related to reservoir compaction and seabed subsidence, as a result of the gas production.

Currently PETRONAS is planning to use the field for CO_2 injection and storage from nearby high CO_2 content gas producing fields. It is therefore important to understand the geomechanics-related risks such as subsidence and uneven lateral movement, fault reactivation, and cap rock integrity associated with the long term CO_2 storage design strategy.

A 4-D coupled reservoir geomechanical study has been conducted to model the pore collapse behavior of the carbonate reservoir in the field, calibrate the model with the field measured data, forecast the long term seabed subsidence and lateral movement, and also to study the risks associated with CO_2 injection with long term storage.

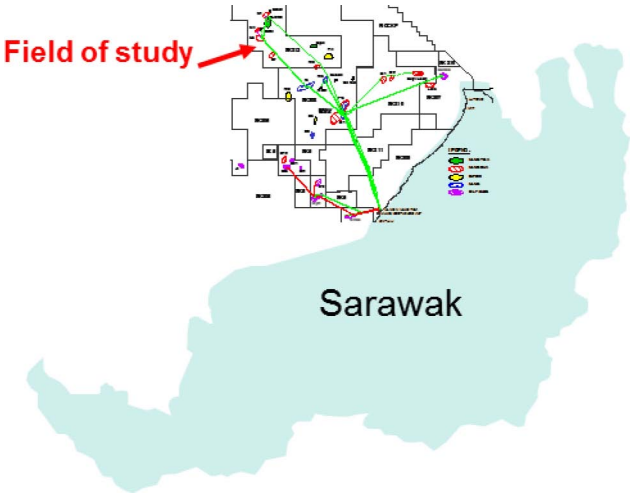


Figure 1—Study field location in offshore Sarawak, Malaysia

Methodology

In order to better delineate and enumerate the risks of CO_2 sequestration, a novel and inclusive workflow (Figure 2) is implemented which includes constructing validated 1-D geomechanical models together with comprehensive laboratory testing programme, incorporating specialized creep tests. The validated 1-D models were combined with structural model and reservoir model to create a field wide 4-D geomechanical model using advanced time-lapsed coupled geomechanical simulations. The finite element simulations coupled with the dynamic reservoir model provided predictions of the caprock seal integrity and fault behavior for the proposed injection scenario by considering shear damage and plastic strains of the seal and faults.

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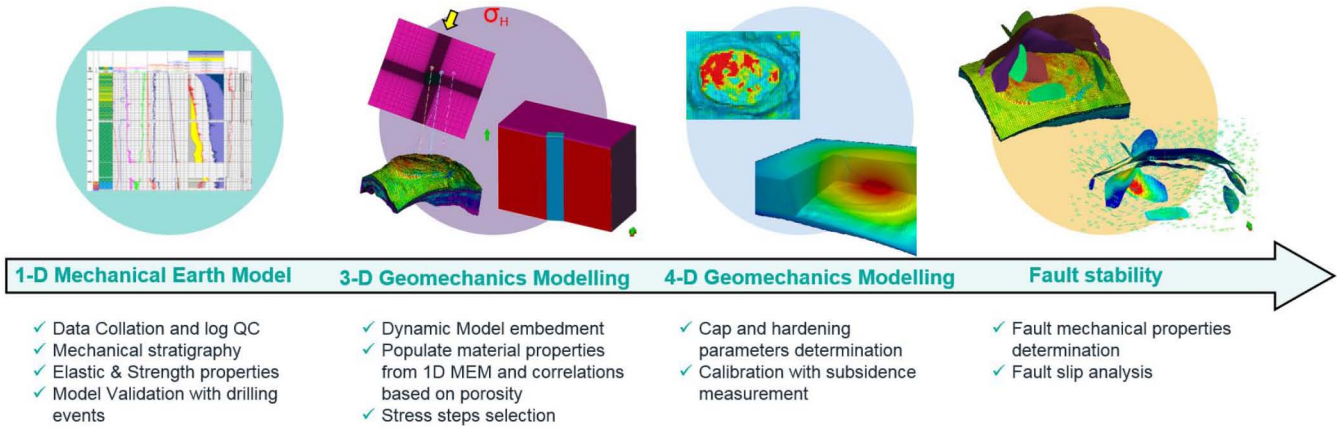


Figure 2—Workflow for 4-D coupled reservoir geomechanical study

1-D Mechanical Earth Model

1-D Mechanical Earth Model (1-D MEM) consists of 3 main components which includes in-situ stresses, rock mechanical properties and pore pressure along the well. It contains all the relevant geomechanics information for any geomechanics-related applications such as wellbore stability analysis, sand production prediction, caprock integrity assessment, fault stability assessment and compaction and subsidence evaluation. Well calibrated 1-D MEMs are essential as a solid foundation for robust geomechanics analyses. Good quality MEM should be able to explain observed drilling experiences with the modelling results.

Four validated 1-D MEMs were selected for the 3-D model construction namely X-1, X-2, X-3 and X-4. The selection criteria are mainly based on data availability (sonic log) and log quality. Rock elastic properties were derived based on sonic logs and conversion from dynamic to static elastic properties was conducted to achieve good match with rock mechanics test data. Rock strength properties were derived from in-house correlations of the region. Emphasis is placed on lithology identification whereby different correlations were applied to different lithologies and to match the wellbore stability predictions with the drilling experiences.

Poroelastic horizontal strain method was used to predict and capture the variability of horizontal stresses in different lithologies and formation. The minimum horizontal stress magnitude was calibrated against leak-off test (LOT) data while the stress azimuth was inferred from compilation of regional stress map in Sarawak Basin (Tjia, 1994). Stress regime understanding from the region is normal stress regime transitional to strike slip regime near top of carbonate and in the carbonate section. Horizontal stress anisotropy is small which is consistent with regional knowledge.

These data are then used for the following in constructing 3-D geomechanical model:

- Derive rock mechanical elastic and strength property correlations with porosity for population in reservoir.
- Determine control points for tectonic stress boundary condition used for 3-D stress initialization.
- Populate overburden rock properties.
- Populate overburden pore pressure (pore pressure in the reservoir is from dynamic model).

One of the 1-D MEMs is shown in Figure 3. Log track from left to right are lithology code (sand, shale and carbonate), well inclination and azimuth, rock elastic properties, rock strength properties, pore pressure and total principal in-situ stresses, mud weight window gradient, bit size and caliper, and casing shoe depth.

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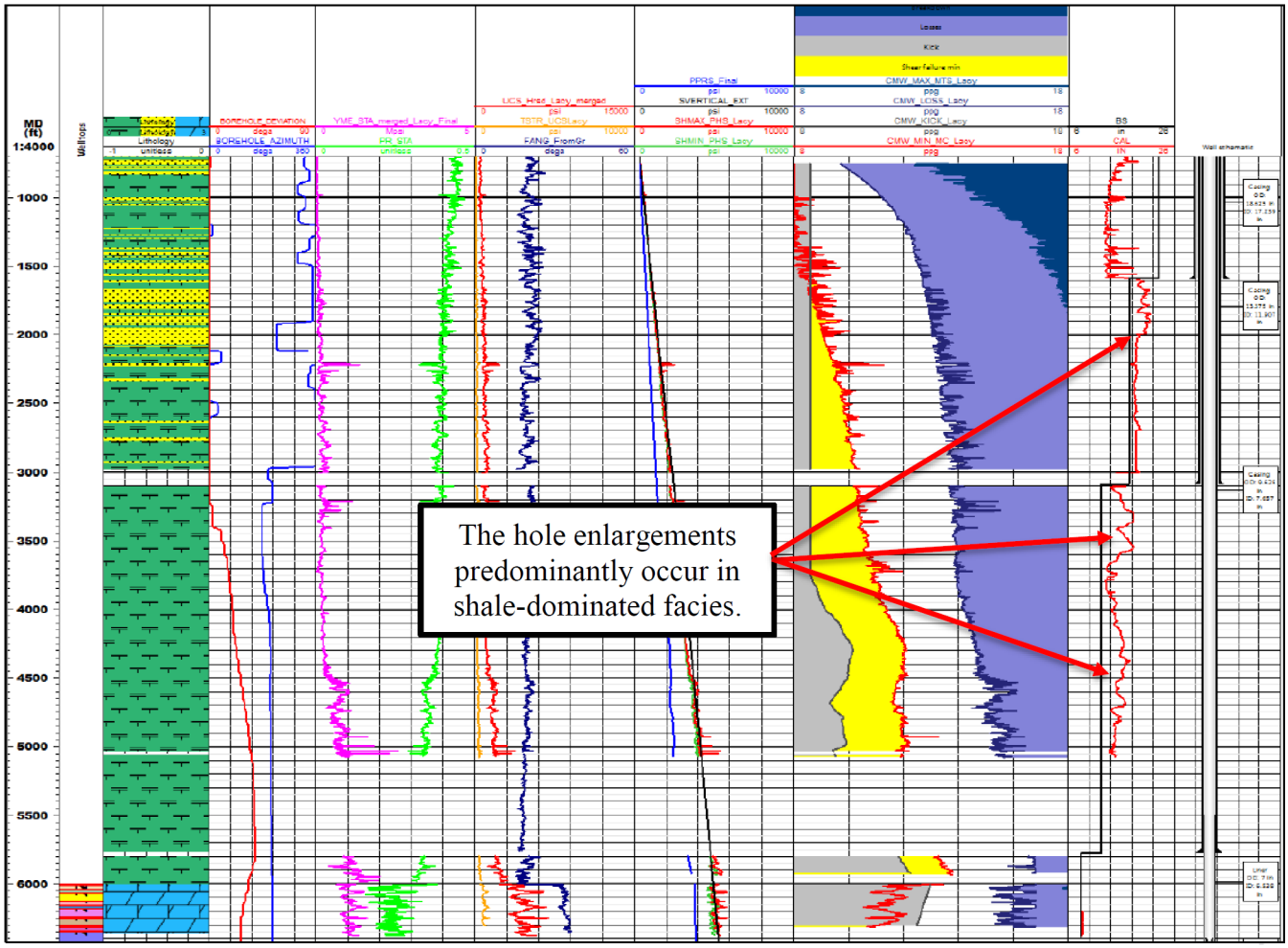


Figure 3—One of the offset well 1-D MEM in Field X

3-D Geomechanical Model

The dynamic reservoir model was used as the basis for construction of the 3-D geomechanical model. The geomechanical model in the reservoir and aquifer zones was extended to seabed (overburden), sideburden and underburden (Figure 4). Extra effort was made to ensure smooth transition between reservoir grid and embedded grid to assist in subsequent numerical computation (Figure 5). The boundary condition was set orthogonal to the maximum horizontal stress direction (20 degrees from North). The total number of grid cells of the embedded 3-D geomechanical model is approximately 2 million cells.

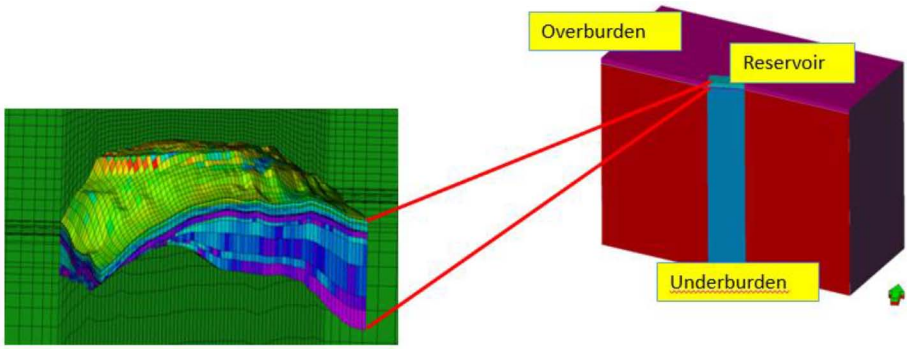


Figure 4—Embedment grid with overburden, underburden and sideburden

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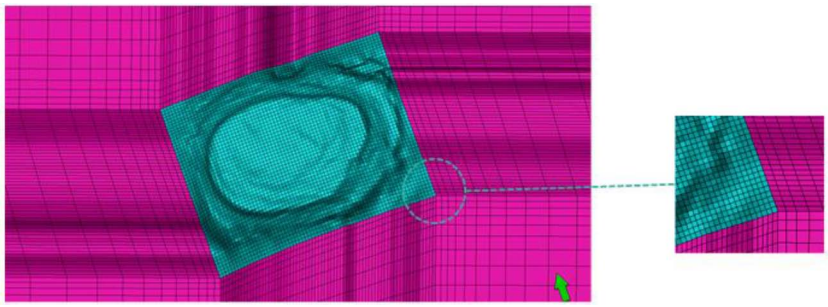


Figure 5—Progression of cell resolution from reservoir to far field

Rock Mechanical Properties Population

The 3-D distribution of density, elastic properties and rock strength properties in the reservoir section were based on porosity correlations as shown in Table 1. It is observed the rock exhibits pore collapse failure during pore pressure depletion in uniaxial pore volume compressibility (UPVC) tests based onp-q plot and porosity vs. volumetric strain plot. Thus, ISAMGEO chalk model (Plischke, 1994) incorporating creeping is used for this field to cater for the cap failure and creeping behavior (evidence from GPS). The model parameters are established based on numerical simulation of UPVC tests with actual laboratory test stress path and subsequently, correlations were derived against sample porosity (Table 2).

Table 1—Rock mechanical properties correlations for reservoir based on calibrated 1-D MEMs data

Rock Mechanical Properties	Field Correlations
Bulk Density (g/cc)	RHOB = 2.6658 - 1.315Phi
Young's Modulus (Mpsi)	YME = 3.478 x exp(-3.281Phi)
Poisson's Ratio (unitless)	PR = 0.33905 - 0.3861Phi
Unconfined Compressive Strength (psi)	UCS = 10598 x exp(-4.5206Phi)
Friction Angle (deg.)	FANG = 33.861 – 3.816 Phi
Tensile Strength (psi)	TSTR = UCS / 10
Dilation Angle (deg.)	DANG = FANG / 2

Table 2—ISAMGEO parameters correlations based on UPVC test data

ISAMGEO Parameters	Field Correlations
Pre-consolidation Pressure (Psi)	Pc_exp=1.8354E+4*exp(-5.541E-2*Phi)
Hardening Parameter (unitless)	H_exp=1.8613E+7*exp(-1.578E-01*Phi)
Plastic Volumetric Strain Reference Rate (unitless)	Rat0 = -0.861 ln(Phi) + 2.8971
Exponent for Rate Dependent Pore Collapse Strength (unitless)	BRat0 = 0.0072 exp(0.0323 Phi)

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The elastic and strength properties in the overburden were populated using kriging method based on the four 1-D MEMs properties. After having populated the 3-D geomechanical model with the rock mechanical properties, quality check was carried out by comparing the 3-D mechanical properties with those of the1-D MEMs. Figure 6 shows the comparison between calibrated3-D geomechanical properties and the 1-D MEMs properties which demonstrate good agreement between the 3-D populated data and field data from the 1-D MEMs.

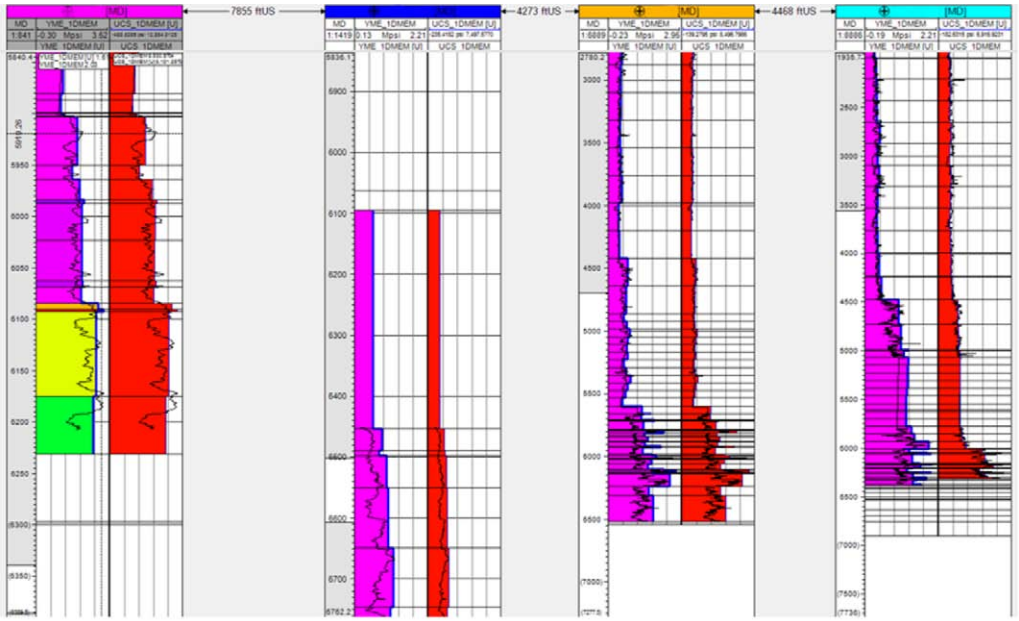


Figure 6—Comparison between mapped 3-D rock mechanical properties (Young's modulus and UCS) against 1-D MEMs properties

Stress Boundary Condition initialization

Initialization of in-situ stresses is inferred from 1-D Mechanical Earth Model whereby iterative boundary conditions were applied until reasonable match between 3-D geomechanical model and 1-D MEM stresses is achieved. The calculation of a representative pre-production stress distribution relied on the consideration of gravity and pore pressure forces acting internally within the model as well as an external horizontal loading scheme acting on the vertical model boundaries.

The stress boundary condition of the model is tabulated in Table 3.

The maximum horizontal stress orientation adopted for this field is N20°E based on the stress orientation of nearby fields and detailed study conducted in Sarawak Basin (Tjia, 1994).

Table 3—Stress boundary condition

Parameters	Values
Sh Gradient	0.78psi/ft
SH/Sh	1.04
Sh Azimuth	110°
Sea Water Pressure Gradient	0.45psi/ft

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Stress-step Selection

The reservoir pressure model at pre-production and subsequent stress-steps were selected based on average field reservoir pressure of history-matched dynamic model. The reservoir pressure data at the selected time-steps were used as inputs to compute the stresses and strains in the reservoir and surrounding formations. Phase 1 represents the production phase, Phase 2 represents the end of production period (increase in pressure is due to strong aquifer drive), and Phase 3 represents the beginning of CO_2 injection period. Figure 7 shows a total of 23 selected stress-steps for the study. Emphasis (more stress-steps) were selected during reservoir depletion and aquifer support phase as GPS measurement data were available during this period whereby a smoother subsidence can be captured for comparison with the GPS data.

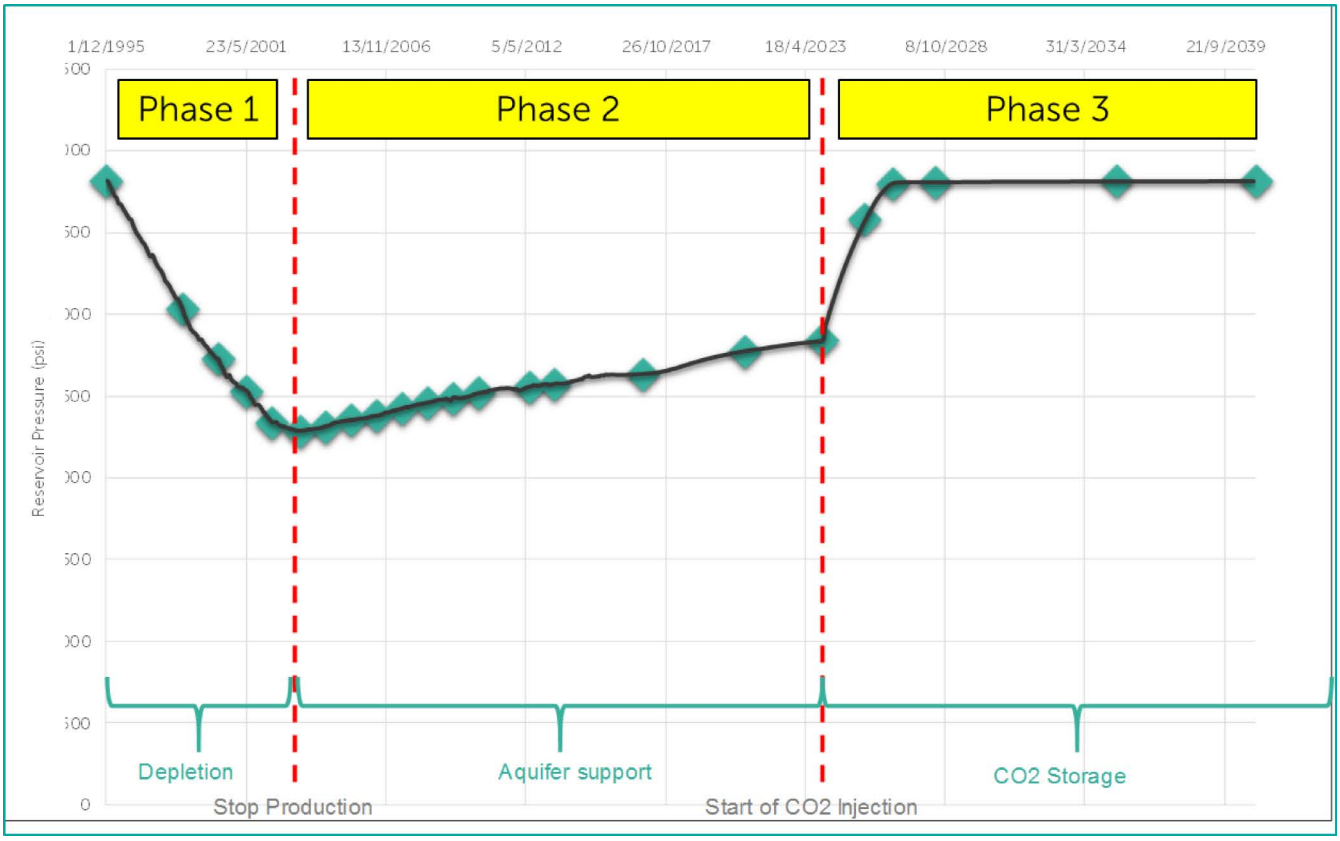


Figure 7—Stress-steps selection

Subsidence and reservoir compaction from depletion

At the end of depletion in Year 2003, seabed subsidence is predicted to be approximately 9.6ft, which then continues due to creep effect. The maximum subsidence predicted is approximately 11.3ft in Year 2013 which is consistent with GPS measurement in 2014 of approximately 11.65ft. Once CO_2 injection is initiated in Year 2024, the reservoir experienced additional pressure support which resulted in seabed uplift of approximately 1ft. At the end of the injection scenario, a maximum seabed subsidence of 10.6ft and reservoir compaction of 16ft were predicted as shown in Figure 9. Iterative calibration of pore collapse and creep parameters from laboratory test data range was conducted to achieve reasonable to good match against subsidence from GPS measurements of platform. Figure 10 shows the aerial view comparison of seabed bathymetry survey from 2002 and 2012 against the model predictions. The subsidence bowl was observed to be qualitatively consistent at North-West of the platform location.

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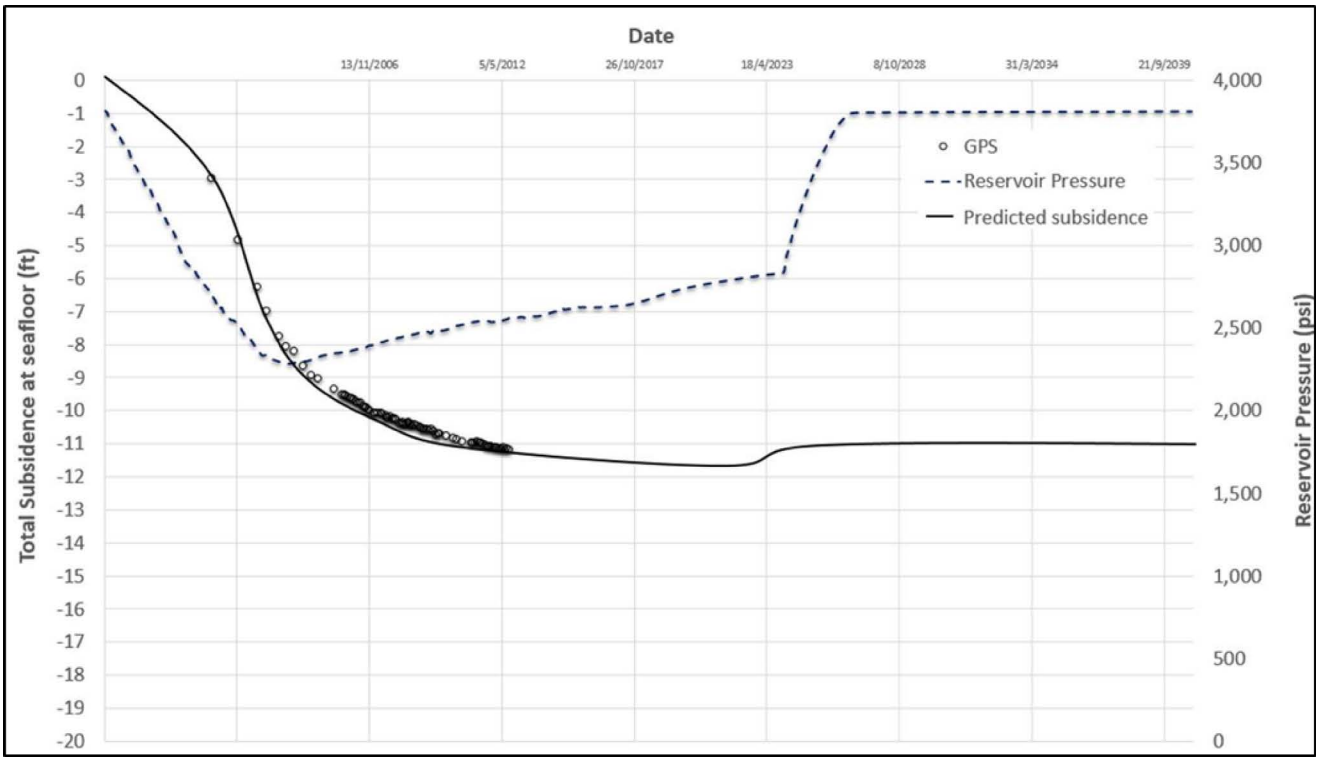


Figure 8—Seabed subsidence and GPS elevation measurements at platform location plotted against time.

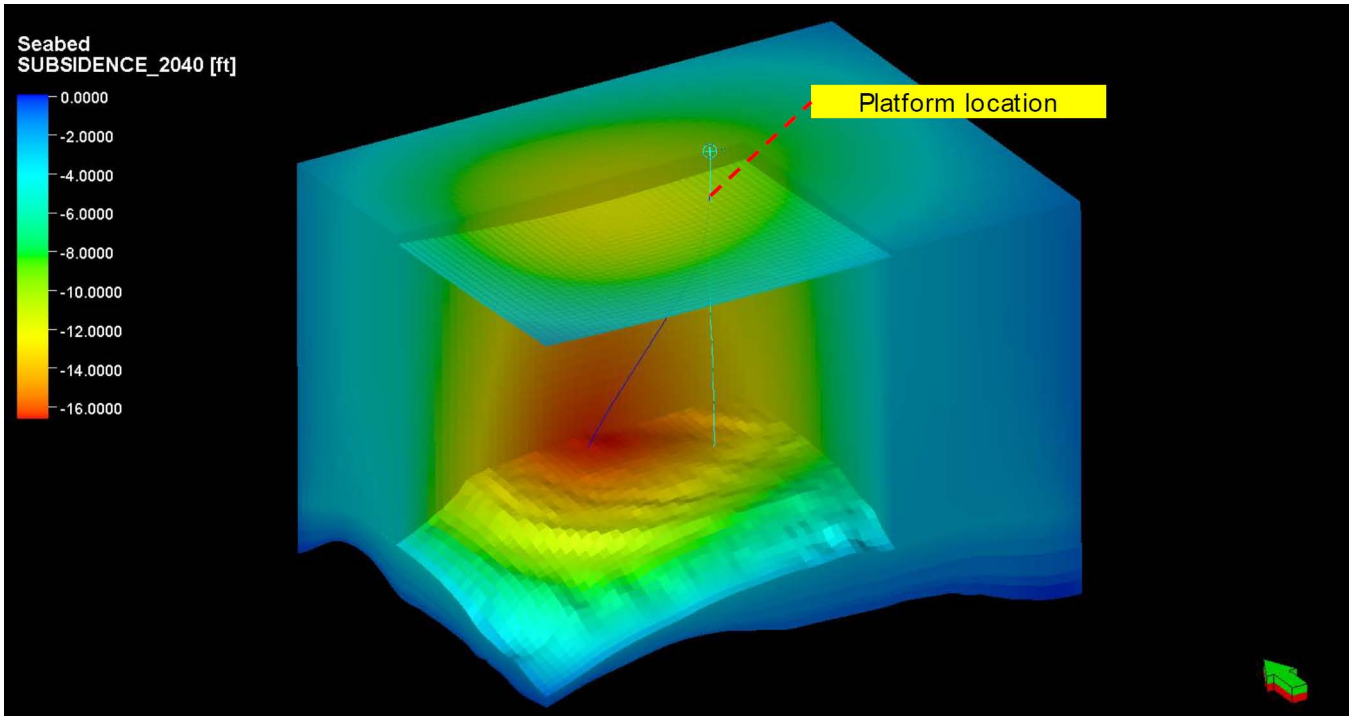


Figure 9—Predicted reservoir compaction and seabed subsidence at Year 2040

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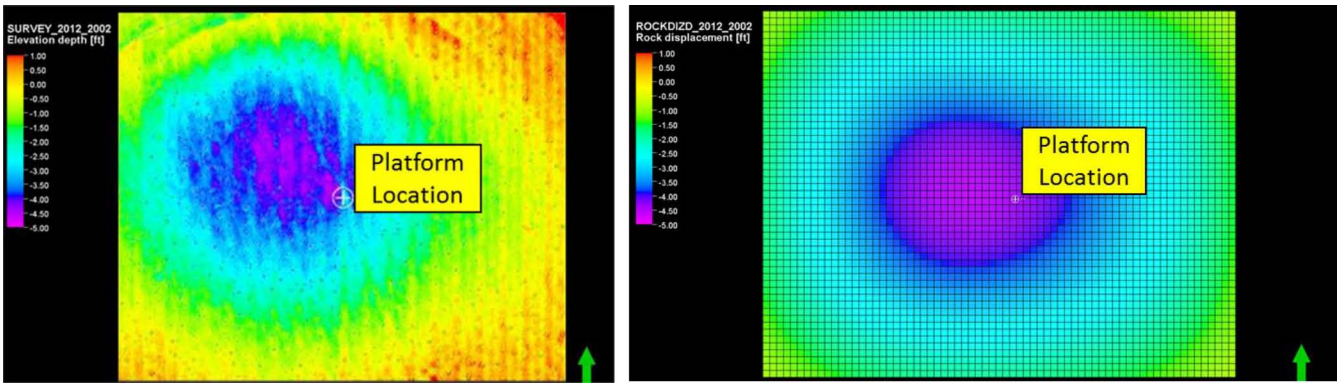


Figure 10—Comparison between measured seabed subsidence from bathymetry survey (left) and the predicted subsidence (right).

Fault Stability Analysis

Numerical fault stability analysis utilizing frictional element is among one of the approaches adopted in the industry. With the use of critical state failure model (ISAMGEO), plastic strain is unable to be computed along the frictional elements (Mohr Coulomb failure criteria). Therefore, a workaround is implemented whereby the results of fault deformation along the fault surface were interpreted to assess the fault reactivation potential.

Another challenge in fault stability modeling is the lack of fault mechanical properties as rock sample is not commonly taken from a fault. In order to estimate suitable fault mechanical properties, a literature review was conducted as summarized in Table 4. From this investigation, a wide range of fault mechanical properties has been used in different fields for fault stability study. In this study, the approach and fault properties discussed by Khodarverdian, et al. (2014) was adopted as it was conducted in offshore Sabah which is regionally close to Sarawak where the field is located.

Table 4—Fault Properties from literature

Parameter	Makurat, et al (1995)	Pereira, et al (2002)	Mendes, et al (2010)	Khodaverdian, et al (2014)
Field Name	Ekofisk Field, North Sea	Synthetic Reservoir	Miranga Field, Brazil	SF-30, Sabah, Malaysia
Type of Reservoir	Carbonate (Chalk)	Data from Literature (Petrobras)	Clastic	Clastic
Fault Normal Stiffness, psi/ft	1,900,800	-	459,600	66,311
Fault Shear Stiffness, psi/ft	530,400	-	171,600	30,945
Fault Cohesion, psi	2465	72.5	110.2	217.5
Fault Friction Angle, deg	33	20	18.3	15

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Figure 11 shows the results of fault deformation analysis. From a total of 12 faults located in the overburden, the model predicted various degrees of deformation and the deformation approximately 0.5cm, are identified in Fault 1, Fault 5, Fault 7 and Fault 10 which are shown in Figure 12 at different stress-steps.

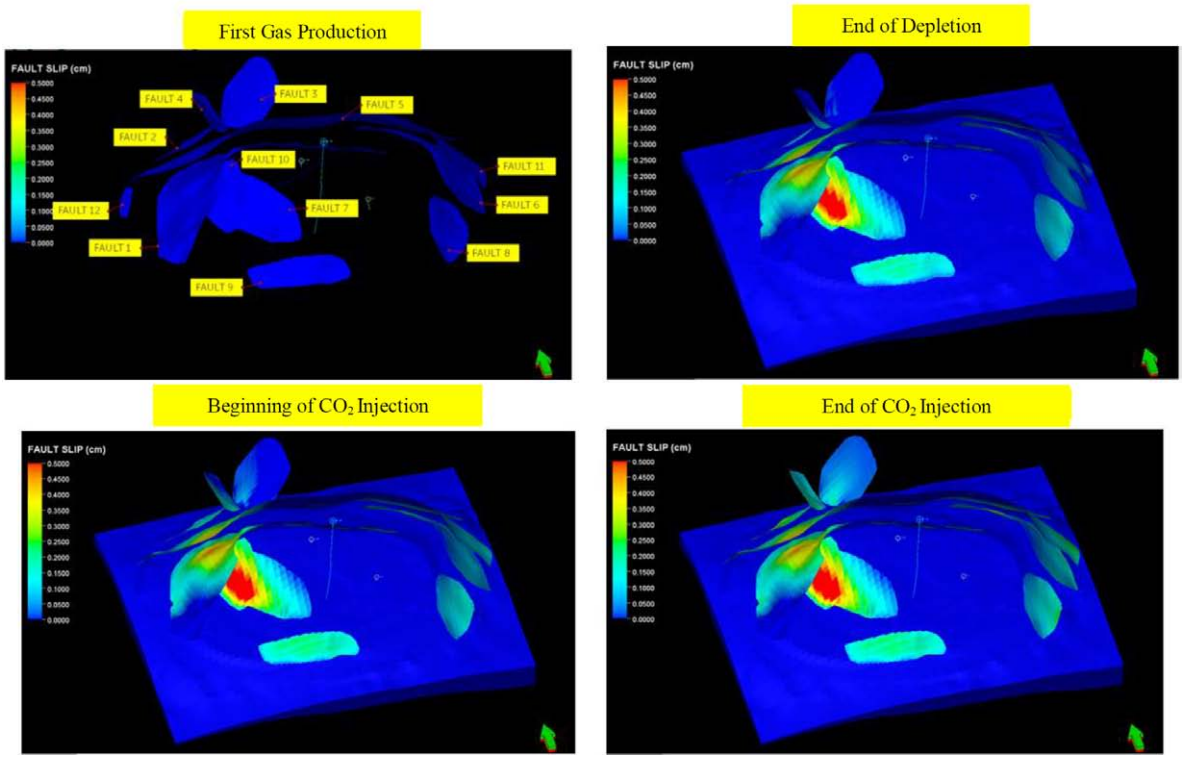


Figure 11—Fault deformation at different stress-steps

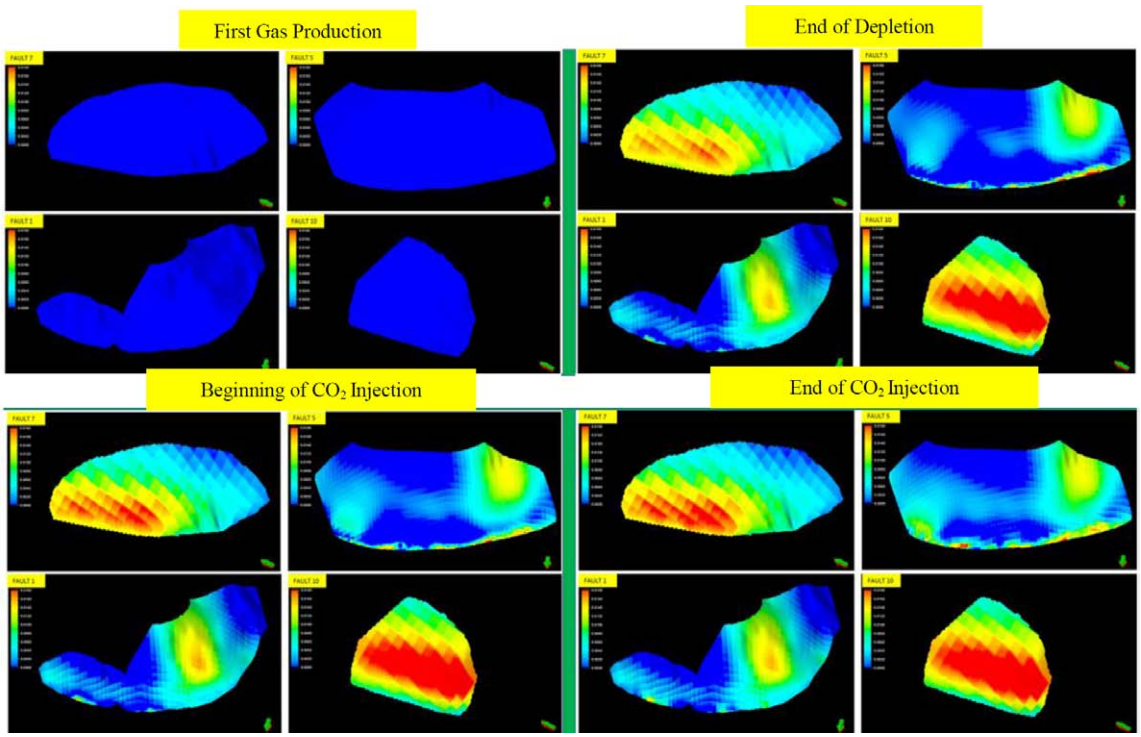


Figure 12—Fault deformation at Fault 1, 5, 7 and 10 at different stress-steps

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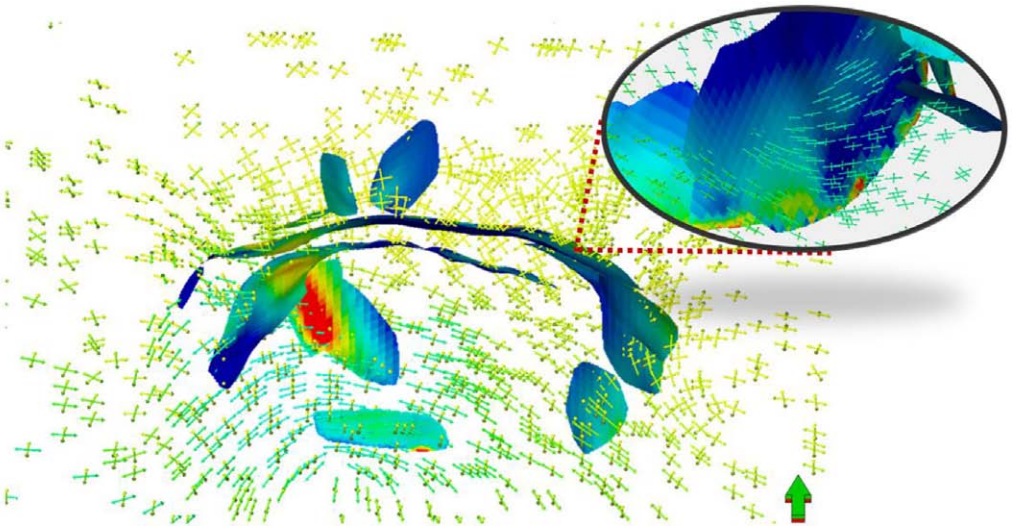


Figure 13—Stress tensor rotation in fault vicinity which is critical for well trajectory optimization with respect to wellbore stability

The fault stability analysis in Field X predicted maximum deformation of approximately 0.5cm throughout the production and CO_2 injection scheme. The fault deformation mainly occurred during depletion phase and comparison is made with predicted deformation from other published fault stability analysis in Table 12. The published findings are consistent with the field experience in Field X whereby there is no reported well casing problem from any gas producers in the fault vicinity during the production period. The study results show that the CO_2 injection contributes minimally to the fault deformation. However, the potential leakage pathway of the cement bond behind casing and cap rock is not included in the study.

Table 5—Qualitative comparison of predicted fault slippage against published data

Parameter	Makurat, et al (1995)	Khodaverdian, et al (2014)	Field X
Field Name	Ekofisk Field, North Sea	SF-30, Sabah Malaysia	Field X, Sarawak, Malaysia
Type of Reservoir	Carbonate (Chalk)	Clastic	Carbonate
Location of Faults	Faults within reservoir extended to overburden and faults above reservoir	Faults within reservoir extended to overburden	Faults in overburden above reservoir
Predicted Maximum Fault Slippage (cm)	43cm (near drawdown area)	5.9cm	0.5cm
Impact of Fault Slippage/Deformation	Observed well casing problem	No impact on casing/completions	No potential impact on casing/completions

Caprock Integrity

During reservoir depletion phase, the pressure decrease will in turn increase the effective stresses within the reservoir and also change the total lateral (horizontal) stresses within and around the reservoir due to poro-elastic effect. The opposite applies for injection scenario. Plastic deformation was computed throughout all the stress-steps. Figure 14 shows the developed plastic strains mainly during depletion phase. Plastic

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strain development is minimal during CO_2 Injection period. The highest plastic strain in the model is approximately 2% and those cells are located at the reservoir crest where the porosity is high(>30%).

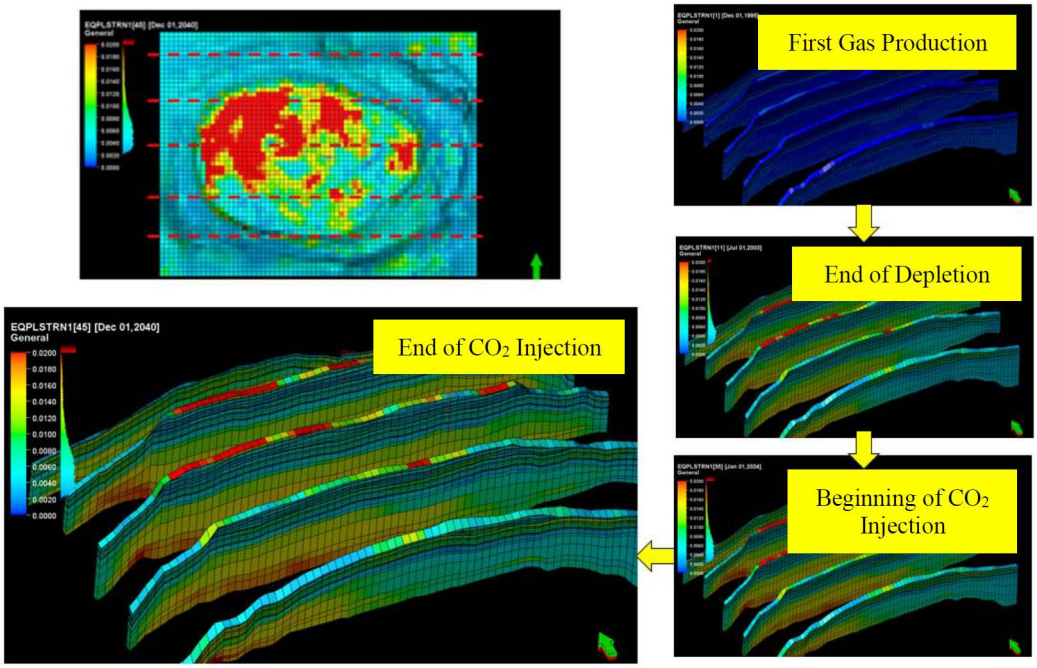


Figure 14—Cross-section of plastic strain distribution. High plastic strain cells are observed at the top layer whereby the porosity is more than 30%.

The severity of plastic strain was compared with other field studies where failure has occurred. In Table 6, the two fields observed some geomechanics-related failures when the average plastic strain is greater than 2% plastic strain. However, no geomechanics-related failure (casing buckling, excessive solid production etc.) was observed in Field X to date. Note that majority of the plastic strain was generated during depletion phase and there is minimal incremental plastic strain during CO_2 injection phase. Hence the potential of geomechanics-related failures during CO_2 injection phase is low.

Table 6—Plastic strain comparison with geomechanic failures in other published geomechanical studies

Field	Average Reservoir Plastic Strain (Compaction) at Point Failure	Geomechanic Failure Observed	SPE Reference
Ekofisk (Chalk) North Sea	2.0-3.5%	Casing shear in overburden within 457m of reservoir at sand/shale interface (slip planes). Column buckling within reservoir.	SPE 17856
Matagorda (Sandstone) Texas, USA	2.1-3.25%	Parted production casings at 3.25% average reservoir compaction strain. Excessive sand production from cased and perforated completions at 2.0-3.0% average reservoir compaction strain. Excessive sand production from cased hole gravel pack integrity failure between 2-2.3% average reservoir compaction strain. Catastrophic completion failure of Phase II wells after 0.25-1.25% incremental average reservoir compaction strain.	SPE 84553

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Hydraulic fracture or sub-vertical fracture can propagate within the reservoir and cap rock when the injection pressure is greater than Shmin. Figure 15 shows the evolution of minimum horizontal stress in the field. Shmin has only small changes and remained within 4600psi to 6200psi during CO_2 injection phase. Thus it can be concluded with the proposed CO_2 injection scheme, the field will experience low risk of hydraulic fracture and maximum allowable injection pressure should be limited to 4000psi with 600psi safety margin.

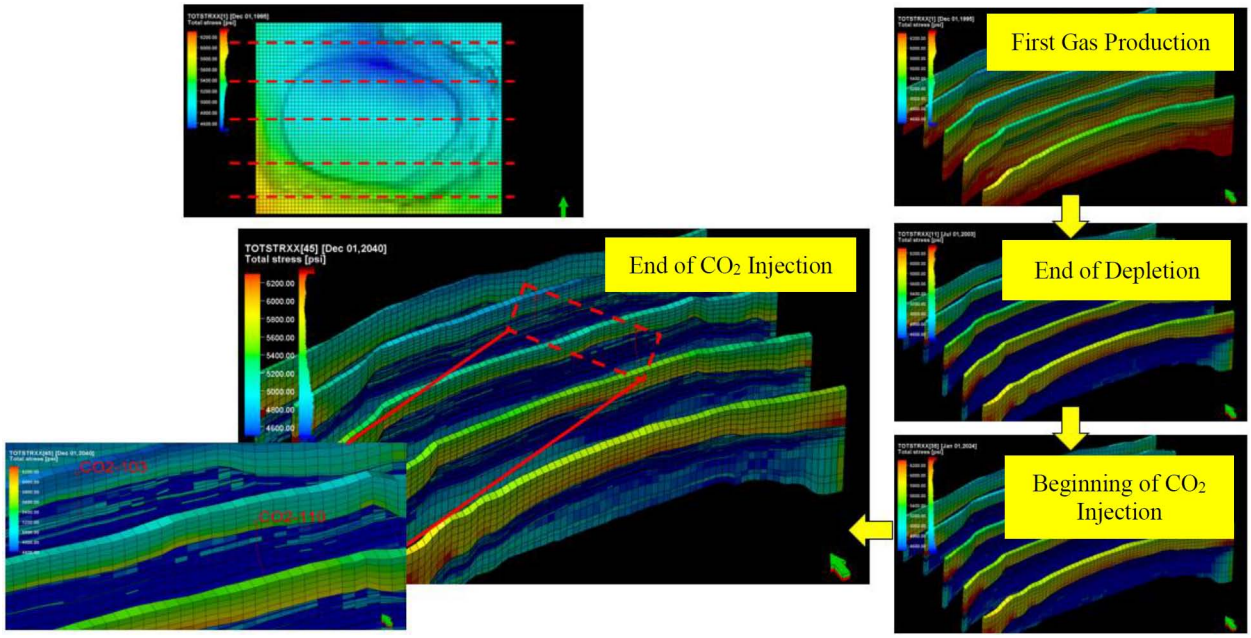


Figure 15—Cross-section of 4-D geomechanical model showing Shmin remained between 4600psi to 6200psi during CO_2 injection phase

Conclusions

The feasibility study on CO_2 injection and storage in Field X using 4-D coupled geomechanical modelling generates the following findings:

- Comparison of seabed subsidence data obtained from GPS measurements up to 2014 with predicted subsidence are in good agreement.
- The modelling suggests 10.6ft of seabed subsidence and 16ft of reservoir compaction at the end of injection scenario in Year 2040.
- Localized reservoir rock failure based on plastic strain was observed at the crest of the reservoir.
- Current depletion and proposed injection scheme could result in limited fault deformation in the order of 0.5cm at five overburden bounding faults.
- The recommended maximum allowable injection reservoir pressure is 4000psi with 600psi extended safety margin in order to minimize the risk for generation of sub-vertical fracture at the cap rock considering stress changes and plastic strains based on compaction, subsidence, cap rock integrity and fault stability analyses.

The study offers the utilization of the latest techniques in 4-D coupled geomechanical modelling hence reduced the study time and costs significantly, making it affordable for in-time solutions for decision making. The paper also aims to encourage the community and industry to consider the workflow involved in CCS strategy evaluation, focusing on risk assessment which ultimately will affect reservoir maximum safe

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injection limit, storage capacity, long term storage safety, and monitoring programme to mitigate potential geohazard leakage.

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